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Substrate structures and methods of manufacture

Abstract

A power electronic substrate includes a metallic baseplate having a first and second surface opposing each other. An electrically insulative layer also has first and second surfaces opposing each other, its first surface coupled to the second surface of the metallic baseplate. A plurality of metallic traces each include first and second surfaces opposing each other, their first surfaces coupled to the second surface of the electrically insulative layer. At least one of the metallic traces has a thickness measured along a direction perpendicular to the second surface of the metallic baseplate that is greater than a thickness of another one of the metallic traces also measured along a direction perpendicular to the second surface of the metallic baseplate. In implementations the electrically insulative layer is an epoxy or a ceramic material. In implementations the metallic traces are copper and are plated with a nickel layer at their second surfaces.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of the earlier U.S. Utility patent application to Lin et al. entitled “Substrate Structures and Methods of Manufacture,” application Ser. No. 15/868,747, filed Jan. 11, 2018, which application is a divisional application of the earlier U.S. Utility patent application to Lin et al. entitled “Substrate Structures and Methods of Manufacture,” application Ser. No. 15/206,574, filed Jul. 11, 2016, now issued as U.S. Pat. No. 9,883,595, which application is a divisional application of the earlier U.S. Utility patent to Lin et al. entitled “Substrate Structures and Methods of Manufacture,” application Ser. No. 14/534,482, now issued as U.S. Pat. No. 9,408,301, the disclosures of each of which are hereby incorporated entirely herein by reference.

BACKGROUND

1. Technical Field

(1) Aspects of this document relate generally to substrate structures for semiconductor integrated circuit components. More specific implementations involve substrate structures for power modules.

2. Background Art

(2) Substrate structures for semiconductor integrated circuits, such as power modules, are used to route components internal and external to an integrated circuit and to dissipate heat. Direct bonded copper (DBC) substrates include a ceramic layer with a layer of copper bonded to one or both sides. Insulated metal substrate (IMS) substrates include a metal baseplate covered by a thin layer of dielectric (usually an epoxy-based layer) and a layer of copper.

SUMMARY

(3) Implementations of power electronic substrates may include: a metallic baseplate having a first surface and a second surface opposing the first surface; an electrically insulative layer having a first surface coupled to the second surface of the metallic baseplate, the electrically insulative layer having a second surface opposing the first surface of the electrically insulative layer; a plurality of metallic traces, each metallic trace coupled to the second surface of the electrically insulative layer at a first surface of the metallic trace, each metallic trace having a second surface opposing the first surface of the metallic trace; wherein at least one of the metallic traces has a thickness measured along a direction perpendicular to the second surface of the metallic baseplate that is greater than a thickness of another one of the metallic traces also measured along a direction perpendicular to the second surface of the metallic baseplate.

(4) Implementations of power electronic substrates may include one, all, or any of the following:

(5) At least one of the metallic traces may have two thicknesses of different magnitudes measured along a direction perpendicular to the second surface of the metallic baseplate.

(6) Each of the plurality of metallic traces may include copper.

(7) Each of the metallic traces may include a layer of nickel plated onto the second surface of the metallic trace.

(8) The electrically insulative layer may include an epoxy.

(9) Implementations of methods of forming an insulated metal substrate (IMS) for a power electronic may include: partially etching a first surface of a copper layer to form a pattern having a first thickness and a second thickness greater than the first thickness, the first thickness and the

second thickness both measured perpendicular to a second surface of the copper layer opposite the first surface of the copper layer; laminating the first surface of the copper layer with a second surface of a dielectric layer, the dielectric layer coupled to a metallic baseplate at a first surface of the dielectric layer opposite the second surface of the dielectric layer and at a second surface of the metallic baseplate; and forming traces in the copper layer by etching through the copper layer at the first thickness and etching through the copper layer at the second thickness, wherein the traces have two different trace thicknesses, where the trace thicknesses are measured perpendicularly to the first surface of the dielectric layer.

(10) Implementations of methods of forming an IMS for a power electronic may include one, all, or any of the following:

(11) Forming the pattern may further include forming a third thickness different in magnitude from the first thickness and the second thickness, the third thickness measured perpendicular to the first surface of the dielectric layer.

(12) Plating a layer of nickel onto the second surface of the copper layer, wherein forming the traces in the copper layer further includes etching through the layer of nickel.

(13) Etching through the copper layer at the first thickness and etching through the copper layer at the second thickness may include coupling a first layer of photoresist to the second surface of the copper layer, forming a pattern in the first layer of photoresist, etching through the copper layer at the first thickness at spaces in the pattern in the first layer of photoresist, removing the first layer of photoresist, coupling a second layer of photoresist to the second surface of the copper layer, forming a pattern in the second layer of photoresist, etching through the copper layer at the second thickness at the spaces in the pattern in the second layer of photoresist, and removing the second layer of photoresist.

(14) Laminating the traces in the copper layer with a second dielectric layer, coupling a second copper layer to the second dielectric layer, and forming traces in the second copper layer to form a stacked IMS for a power electronic.

(15) Implementations of methods of forming a direct bonded copper (DBC) substrate for a power electronic may include: coupling a first surface of a ceramic layer to a second surface of a metallic baseplate, the ceramic layer having a second surface opposing the first surface; coupling a first surface of a copper layer with the second surface of the ceramic layer, the first surface of the copper layer having a pattern including a first thickness and a second thickness greater than the first thickness, the first thickness and the second thickness both measured perpendicularly to the first surface of the ceramic layer; and forming traces in the copper layer by etching through the copper layer at the first thickness and etching through the copper layer at the second thickness, wherein the traces include two different trace thicknesses, where the trace thicknesses are measured perpendicularly to the first surface of the ceramic layer.

(16) Implementations of methods of forming a DBC may include one, all, or any of the following:

(17) The second surface of the ceramic layer may include a pattern complementary to the pattern of the first surface of the copper layer.

(18) Plating a layer of nickel onto the second surface of the copper layer, wherein forming traces in the copper layer includes etching through the layer of nickel.

(19) Etching through the copper layer at the first thickness and etching through the copper layer at the second thickness may include coupling a first layer of photoresist to the second surface of the copper layer, forming a pattern in the first layer of photoresist, etching the copper layer at the first thickness at a space in the pattern in the first layer of photoresist, removing the first layer of photoresist, coupling a second layer of photoresist to the second surface of the copper layer, forming a pattern in the second layer of photoresist, etching the copper layer at the second thickness at a space in the pattern in the second layer of photoresist, and removing the second layer of photoresist.

(20) Implementations of power electronic substrates may include: a dielectric layer having a first

surface and a second surface opposing the first surface; a ceramic layer having a bonding pattern on a second surface of the ceramic layer, the second surface of the ceramic layer laminated with the first surface of the dielectric layer, the first surface of the dielectric layer configured to receive the bonding pattern of the second surface of the ceramic layer; and a plurality of copper traces, each copper trace coupled to the dielectric layer at the second surface of the dielectric layer and at a first surface of the copper trace.

(21) The dielectric layer may include an epoxy.

(22) Implementations of power electronic substrates may include one, all, or any of the following:

(23) A first surface of the ceramic layer may include a bonding pattern, the first surface of the ceramic layer may be laminated with a second surface of a second dielectric layer, the second surface of the second dielectric layer may be configured to receive the bonding pattern of the first surface of the ceramic layer, the power electronic substrate may further have a metallic baseplate having a first surface and a second surface, the second surface of the metallic baseplate being opposite the first surface of the metallic baseplate, the second surface of the metallic baseplate being coupled to a first surface of the second dielectric layer opposing the second surface of the second dielectric layer.

(24) The bonding pattern of the second surface of the ceramic layer may include bonding ridges.

(25) The bonding pattern of the second surface of the ceramic layer may include one of conical and pyramidal projections.

(26) The plurality of copper traces may include at least two trace thicknesses of different magnitudes measured along a direction substantially perpendicular to the second surface of the metallic baseplate.

(27) The foregoing and other aspects, features, and advantages will be apparent to those artisans of ordinary skill in the art from the DESCRIPTION and DRAWINGS, and from the CLAIMS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Implementations will hereinafter be described in conjunction with the appended drawings, where like designations denote like elements, and:

(2) FIG. 1 is a cross-section view of an implementation of an insulated metal substrate (IMS);

(3) FIG. 2 is a cross-section view of another implementation of an IMS;

(4) FIG. 3 is a cross-section view of an implementation of a direct bonded copper (DBC) substrate;

(5) FIG. 4 is a cross-section view of another implementation of a DBC substrate;

(6) FIG. 5 is a cross-section view of a copper layer having photoresist layers thereon

(7) FIG. 6 is a cross-section view of the elements of FIG. 5 with a pattern formed in one of the photoresist layers;

(8) FIG. 7 is a cross-section view of the elements of FIG. 6 with a pattern etched into the copper layer;

(9) FIG. 8 is a cross-section view of the copper layer of FIG. 7 with the photoresist layers removed;

(10) FIG. 9 is a cross-section view of the copper layer of FIG. 8, a dielectric layer and a metallic baseplate of an IMS prior to being coupled together;

(11) FIG. 10 is a cross-section view of the elements of FIG. 9 coupled together;

(12) FIG. 11 is a cross-section view of the elements of FIG. 10 with nickel plating atop the copper layer;

(13) FIG. 12 is a cross-section view of the elements of FIG. 11 with a first layer of photoresist placed atop the nickel plating;

(14) FIG. 13 is a cross-section view of the elements of FIG. 12 with a pattern formed in the layer of photoresist;

- (15) FIG. 14 is a cross-section view of the elements of FIG. 12 with the nickel plating and copper layer having been etched through at the pattern in the first layer of photoresist and the first layer of photoresist then removed;
- (16) FIG. 15 is a cross-section view of the elements of FIG. 14 with a second layer of photoresist placed thereon;
- (17) FIG. 16 is a cross-section view of the elements of FIG. 15 with a pattern formed in the second layer of photoresist;
- (18) FIG. 17 is a cross-section view of the elements of FIG. 16 with the nickel plating and copper layer having been etched through at the pattern in the second layer of photoresist and the second layer of photoresist then removed;
- (19) FIG. 18 is a cross-section view of the copper layer of FIG. 8 having a pattern thereon, a ceramic layer having a complementary pattern, and a metallic base plate of a DBC substrate prior to being coupled together;
- (20) FIG. 19 is a cross-section view of the elements of FIG. 18 coupled together;
- (21) FIG. 20 is a cross-section view of the elements of FIG. 19 with a layer of nickel plated onto the copper layer;
- (22) FIG. 21 is a cross-section view of the elements of FIG. 20 with a first layer of photoresist placed atop the nickel plating;
- (23) FIG. 22 is a cross-section view of the elements of FIG. 21 with a pattern formed in the first layer of photoresist;
- (24) FIG. 23 is a cross-section view of the elements of FIG. 22 with the nickel and copper layers having been etched through at the pattern in the first layer of photoresist and the first layer of photoresist having being removed;
- (25) FIG. 24 is a cross-section view of the elements of FIG. 23 with a second layer of photoresist placed thereon;
- (26) FIG. 25 is a cross-section view of the elements of FIG. 24 with a pattern formed in the second layer of photoresist;
- (27) FIG. 26 is a cross-section view of the elements of FIG. 25 with the nickel and copper layers having been etched through at the pattern in the second layer of photoresist and the second layer of photoresist having been removed;
- (28) FIG. 27 is a cross-section close-up view a substrate implementation having a copper layer, a first dielectric layer, a ceramic layer, a second dielectric layer, and a metallic baseplate magnified;
- (29) FIG. 28 is a cross-section view of the elements of FIG. 27 shown with less magnification;
- (30) FIG. 29 is a cross-section view of the copper layer of FIG. 8 having a pattern thereon, a first dielectric layer, a ceramic layer having a pattern complementary to the copper layer, a second dielectric layer, and a metallic baseplate of a power electronic substrate prior to fully coupling the elements together; and
- (31) FIG. 30 is a cross-section view of the elements of FIG. 29 fully coupled together.

DESCRIPTION

(32) This disclosure, its aspects and implementations, are not limited to the specific components, assembly procedures or method elements disclosed herein. Many additional components, assembly procedures and/or method elements known in the art consistent with the intended substrate structures and methods of manufacture will become apparent for use with particular implementations from this disclosure. Accordingly, for example, although particular implementations are disclosed, such implementations and implementing components may comprise any shape, size, style, type, model, version, measurement, concentration, material, quantity, method element, step, and/or the like as is known in the art for such substrate structures and methods of manufacture, and implementing components and methods, consistent with the intended operation and methods.

(33) Referring now to FIG. 1, an implementation of a power electronic substrate 2 is illustrated that

includes an insulated metal substrate (IMS) **4**. The IMS **4** has a metallic baseplate **6** which may be formed of, by non-limiting example, aluminum, copper, steel, and other heat-conducting materials. The metallic baseplate **6** has a first surface **8** which is configured to couple to, by non-limiting example, a heat sink, a motherboard, and the like. The metallic baseplate **6** has a second surface **10** on an opposite (opposing) side from the first surface **8**.

(34) A dielectric layer **12** is coupled to the metallic baseplate **6**. The dielectric layer **12** has a first surface **14** which is coupled to the second surface **10** of the metallic baseplate **6** and a second surface **16** on an opposite side of the dielectric layer **12** from the first surface **14**. In various implementations the dielectric layer **12** includes a resin or epoxy **18**, though in other implementations it may include other dielectric (electrically insulative) materials.

(35) A plurality of traces **20** are formed and coupled to the dielectric layer **12**. Each trace **20** has a first surface **22** coupled to the second surface **16** of the dielectric layer **12** and a first surface **22** on an opposite side of the trace **20** from the first surface **22**. The traces **20** are metallic and may be formed of, by non-limiting example, copper, aluminum, or other electrically conductive materials. Some of the traces **20** have a first thickness **26**, measured from the first surface **22** to the second surface **24**, and some of the traces **20** have a second thickness **28**, greater than the first thickness **26**, measured from the first surface **22** to the second surface **24**. In some implementations there could be traces **20** having a third thickness sized differently from both the first thickness **26** and second thickness **28** or other traces that contain both the first thickness and the second thickness. Referring to FIGS. 7-9, the difference in thicknesses is created at least in part by a pattern **100** which is formed in a first surface **98** of a copper layer **96** from which the traces **20** are formed, which will be discussed hereafter, and the traces **20** which have the smaller first thickness **26** correspond with the pattern **100** or, in other words, are located at the pattern **100** or formed of the material that composes the pattern **100**. Referring back to FIG. 1, a layer of nickel **30** is included on the second surface **24** of each metallic trace **20**. In implementations a single trace **20** may have different thicknesses in different places and so may include the first thickness **26**, second thickness **28**, a third thickness, and so on. A trace **20** of this nature is illustrated in FIG. 17.

(36) Referring now to FIG. 2, in particular implementations a power electronic substrate **32** is an IMS **34** that is similar in structure to IMS **4** except the traces lack the nickel **30** atop the traces **20**.

(37) Referring now to FIGS. 3 and 4, implementations of a power electronic substrate **36** that are direct bonded copper (DBC) substrates are illustrated. The DBC substrate **38** has a metallic baseplate **40** which may be formed of, by non-limiting example, copper, aluminum, steel, and the like. The metallic baseplate **40** has a first surface **42** configured to be coupled to, by non-limiting example, a heat sink, a motherboard, and the like, and further has a second surface **44** on an opposite side of the metallic baseplate **40** from the first surface **42**. A first surface **48** of a ceramic layer **46** is coupled to the second surface **44** of the metallic baseplate **40**. The ceramic layer **46** has a second surface **50** on an opposite side of the ceramic layer **46** from the first surface **48**. A pattern **52** is formed in the second surface **50** of the ceramic layer **46** which may be formed, by non-limiting example, with a number of patterning techniques used to etch and shape ceramic materials. The ceramic layer **46** may be half-etched, though in implementations the etching may go more or less than halfway through the ceramic layer **46**. The etching may be accomplished through wet-etching techniques. In other implementations, the ceramic layer **46** may be patterned through printing, molding, or stamping when the ceramic material is still soft and pliable before curing, firing, or sintering of the layer has taken place.

(38) The DBC substrate **38** has a plurality of traces **20** similar to IMS **4**. The traces **20** having the larger second thickness **28**, measured between the first surface **22** and second surface **24**, correspond with the pattern **52**, or in other words are located at or formed from the pattern **52**. A layer of nickel **30** is placed atop each trace **20**, similar to IMS **4**. which may be plated onto the traces **20**.

(39) Referring now to FIG. 4, in implementations a power electronic substrate **54** is a DBC

substrate **56** that is similar to DBC substrate **38** except it lacks the nickel layer **30**.

(40) Referring now to FIGS. **5-17**, a method of forming an IMS **4** is illustrated. A copper layer **96** is first processed where the copper layer **96** has a first surface **98** and a second surface **102** on an opposite side of the copper layer **96** from the first surface **98**. A layer of photoresist **104** is placed on the first surface **98** and another layer of photoresist **104** is placed on the second surface **102**. A pattern is formed in the photoresist **104** on the first surface **98**, as seen in FIG. **6**. This may be done by exposing a portion of the photoresist **104** to ultraviolet (UV) light or other exposure techniques which make a portion of the photoresist **104** more resistant (or more susceptible) to being removed, and then developing the photoresist **104** with a solution that removes the treated (or untreated) portion to form the pattern.

(41) While only a single part of the pattern is shown, it may be understood that FIG. **6** is a close-up view of only a portion of the elements, and that in reality a pattern of traces and other shapes may be formed in the photoresist **104**. An etching process is then used to etch a pattern **100** into the first surface **98** of the copper layer **96** through the spaces formed in the photoresist **104**. This may be done using any conventional etching mechanisms used to etch copper. The formation of the pattern **100** forms locations of the copper layer **96** that have the first thickness **26** and other locations that have the second thickness **28**, the smaller first thickness **26** corresponding with the patterned areas where the first surface **98** has been etched. It can be seen from FIG. **7** that the etching of the copper layer **96** is a partial etch which does not go all the way through to the second surface **102**. In some implementations, the etching may be half-etched. In other implementations, the pattern **100** may be etched more or less than halfway through the copper layer **96**.

(42) Referring now to FIG. **8**, after the pattern **100** has been etched into the first surface **98** the layers of photoresist **104** are removed. It may be understood that the layer of photoresist **104** that was placed on the second surface **102** is used to prevent the second surface **102** from being etched during the etching process—such as, for instance, in cases where the etching was done with wet etching where the entire copper layer **96** was placed in an etching solution. Any of a wide variety of conventional methods for removing the photoresist **104** (ashing, solvent cleaning, etc.) may be employed in various implementations.

(43) Referring now to FIG. **9**, a metallic baseplate **6** is illustrated that has a first surface **8** and second surface **10** as previously described. A dielectric layer **12** having first surface **14** and second surface **16** is also provided, which in the implementation shown includes an epoxy **18**. The copper layer **96** is positioned so that its first surface **98** faces the second surface **16** of the dielectric layer **12**.

(44) Referring now to FIG. **10**, the copper layer **96**, dielectric layer **12**, and metallic baseplate **6** are illustrated after having been coupled together through a laminating or other pressure bonding process that presses the layers together. During the bonding step, the dielectric layer **12** flows under the pressure forces during this step of assembly and accommodates the pattern **100**, as seen in FIG. **10**, embedding the pattern **100** into the dielectric layer **12**. This bonding/laminating step forms a complementary, or substantially complementary pattern, to the pattern **100** in the dielectric layer **12**.

(45) Referring to FIG. **11**, a layer of nickel **30** is plated or otherwise deposited onto the copper layer **96**. As illustrated in FIGS. **12-14**, a first layer **106** of photoresist **104** is placed atop the nickel **30** and a pattern **108** is formed therein. While only one space of the pattern **108** is shown, it may be understood that this is a close-up view showing only a small portion of the elements, so that in reality a number of patterned areas may be formed in the first layer **106** of photoresist **104**. The nickel plating **30** and copper layer **96** are then fully etched down to the dielectric layer **12** at the pattern **108** and then the first layer **106** of photoresist **104** is removed, as seen in FIG. **14**.

(46) A second layer **110** of photoresist **104** is then coated onto the elements as shown in FIG. **15** and a second pattern **112** is formed therein. Although only a single space of the pattern **112** is shown, it may be understood that a number of spaces may be formed therein. The nickel layer **30**

and copper layer **96** are then fully etched through down to the dielectric layer at the pattern **112** to form the traces **20** and the second layer **110** of photoresist **104** is removed. Some of the traces **20** have the first thickness **26** and some have the second thickness **28**—and in the implementation shown some have both the first thickness **26** and second thickness **28**.

(47) It may be perceived that a slightly modified version of this process may be used to form IMS **34** illustrated in FIG. **2**, wherein the step of adding the nickel **30** is unnecessary and the etching processes to form the traces **20** accordingly do not involve etching through the nickel **30**. It may also be understood that the process could be slightly modified to form traces **20** of more than two thicknesses. By non-limiting example, layers of photoresist **104** could be coated onto the copper layer **96** shown in FIG. **8**, a pattern formed therein, and an etching process may then be used to etch a second pattern into the copper layer **96**, which if etched to a different depth in the copper layer **96**, may be used to form a third thickness in the copper layer **96** different than the first thickness **26** and second thickness **28**. This process could be repeated numerous times to form many thicknesses in the copper layer **96**. This may be done with the second surface **16** of the copper layer **96** remaining flat and, accordingly, the remaining process steps are identical or fairly identical to those described previously.

(48) FIGS. **18-26** show a process of forming DBC substrate **38**, which in some aspects is similar to the process described above for forming IMS **4**, as it involves patterning a layer of copper as was previously described for use in subsequent processing. As A pattern **100** is formed in the first surface **98** of the copper layer **96** as already described in this document. With respect to shaping the ceramic layer, a pattern **52** is formed in the second surface **50** of the ceramic layer **46** which is complementary, or substantially complementary, to the pattern **100** in the copper layer **96**. The pattern **52** in the ceramic layer **46** may be formed using any of a variety of techniques for etching or shaping ceramic materials, including photoresist masking and dry or wet etching, or through stamping/forming processes prior to the ceramic material being cured/dried/fired/sintered. As shown in FIG. **19**, the copper layer **96**, ceramic layer **46** and metallic baseplate **40** are bonded together through a sintering or other similar process used to form intermetallic or other bonding layers between the copper and the ceramic material. A layer of nickel **30** is coupled atop the copper layer **96**, through electroplating or deposition as shown in FIG. **20**, and atop this a first layer **106** of photoresist **104** is added as shown in FIG. **21**. A pattern **108** is formed in the first layer **106**, as shown in FIG. **22**. As described above, although only a single space is formed there may be a plurality of spaces in the pattern **108**. The nickel **30** and copper layer **96** are fully etched through at the gap **108**, revealing the ceramic layer **46**, and the first layer **106** of photoresist **104** is then removed, as illustrated in FIG. **23**.

(49) A second layer **110** of photoresist **104** is then added to the elements as shown in FIG. **24** and a pattern **112** is formed therein, as seen in FIG. **25**. Again, there may be a plurality of spaces composed in the pattern **112**. The nickel layer **30** and copper layer **96** are fully etched through at the pattern down to the ceramic layer **46** to form the traces **20**, and the second layer **110** of photoresist **104** is removed. Some traces **20** have the first thickness **26** and some have the second thickness **28** and, if desired, the process may be used to form some traces **20** having both thicknesses, as illustrated in FIG. **26**. As with other processes described above, there may be more than two trace thicknesses by making slight modifications to the process as described above with respect to the process for forming IMS **4** to shape the copper layer **96**. A process for forming DBC substrate **56** may be similar in many respects to the process for forming DBC substrate **38** except that the nickel plating **30** is not included (and, accordingly, is not etched through).

(50) FIG. **27** illustrates a power electronic substrate **58** that can be considered a hybrid as it has some elements similar to an IMS and some elements similar to a DBC substrate. A metallic baseplate **6** is used, having the first surface **8** and second surface **10** as previously described. There are two dielectric layers **60** and **90**, and a ceramic layer **66** is sandwiched therebetween. The second dielectric layer **90** has a first surface **92**, on an opposite side from a second surface **94**, the first

surface **92** being bonded to the second surface **10** of the metallic baseplate **6**.

(51) A first surface **68** of the ceramic layer **66** has a bonding pattern **70** thereon. This may include bonding ridges **72**, conical projections **74**, pyramidal projections **76**, and the like dispersed on the first surface **68** of the ceramic layer. Other patterns and/or shapes may be employed to increase the surface area and/or the surface interaction between the ceramic layer **66** and the dielectric material. Referring to FIG. **27** (and the page on which the drawing is presented) the bonding pattern **70** may include a series of discrete elements that extend through the surface of the page (such as a grid or array of individual projections when viewed from above) and/or rows extending through the page surface. The second surface **94** of the second dielectric layer **90** receives the bonding pattern **70**. This may be accomplished by the second dielectric layer **90** behaving as a fluid when it is being bonded to the ceramic layer **66** via a laminating or other pressure process inducing localized flow of the dielectric material to effectively form a pattern that is complementary, or substantially complementary, to the bonding pattern **70**. The second dielectric layer **90** may be formed of an epoxy **18**, and the bonding pattern **70** may assist the epoxy **18** to bond sufficiently to the ceramic layer **66**.

(52) A second surface **78** of the ceramic layer **66** opposite the first surface **68** also includes a bonding pattern **80**, which may include any features or characteristics previously described with respect to bonding pattern **70**, and may include bonding ridges **82**, conical projections **84**, pyramidal projections **86**, and the like. Other patterns and/or shapes may be used. The first surface **62** of the first dielectric layer **60** receives the bonding pattern **80** and, accordingly, forms a complementary or substantially complementary pattern on the first surface **62**. The first dielectric layer **60** may have any of the characteristics, features, and so forth of the second dielectric layer **90**. A second surface **64** of the first dielectric layer **60**, opposite the first surface **62**, is bonded to a copper layer **96**.

(53) FIG. **28** is a view of the power electronic substrate **58** shown at a lesser degree of magnification so that the bonding patterns **70**, **80** is not visible. Traces **20** may be formed in the copper layer **96** at this point, in a similar manner as described above with respect to other power electronic substrates. The power electronic substrate **58** may have a copper layer **96** (and accordingly, traces **20**) of a uniform thickness, or the copper layer **96** may have a pattern **100** therein and the ceramic layer **66** may have a pattern **88** therein, as seen in FIGS. **29** and **30**, that is complementary, or substantially complementary, to pattern **100** (and may be formed through etching processes as described herein) so that there will be traces **20** of varying thicknesses, which may be formed using techniques already described with respect to other power electronic substrates herein. In other implementations, however, the traces **20** may be formed without varying thicknesses.

(54) In implementations of power electronic substrates disclosed herein which use an epoxy or resin for the dielectric layer, the dielectric layer may have a thickness from its first surface to its second surface of, or of about, 25 microns to, or to about, 300 microns. The epoxy or resin may include thermally conductive filler particles, such as by non-limiting example SiO₂, Al₂O₃, BN, or the like, dispersed therein. Copper layers described herein may be copper foil and may have, by non-limiting example, thicknesses ranging from, or from about, 18 microns to, or to about 200 microns, or greater. In implementations in which the metallic baseplates are formed of aluminum they may have an alumite and/or anodized aluminum layer on the first and second surfaces. Some metallic baseplates may have, by non-limiting example, a thickness from the first surface to the second surface of, or of about, 1.5 mm.

(55) In implementations herein in which a ceramic layer is used the ceramic layer may include, by non-limiting example, alumina, aluminum nitride, and other high thermally conductive ceramic or composite materials. A copper layer may be directly bonded to a ceramic layer using a high-temperature oxidation process wherein the copper and ceramic are heated to a controlled temperature in a nitrogen atmosphere containing about 30 ppm of oxygen (or about 1.5%

concentration of 02 in atom percentage) to form a copper-oxygen eutectic which bonds both to the copper and to an oxide of the ceramic layer. In implementations the ceramic layer may be Al.sub.2O.sub.3 and a thin layer of copper-aluminum-spinel may bond the copper layer to the ceramic layer. In implementations the ceramic layer may be aluminum nitride and a thin layer of copper-aluminum-nitride may be formed by first oxidizing the surface of the aluminum nitride to form a layer of alumina by high temperature oxidation. In implementations a copper layer may be bonded to a ceramic layer using a sintering process. In particular implementations, the sintering process may involve melting or softening small particles comprised in each of the copper layer and the ceramic layer to bond them with adjacent small particles. By small in this process is meant microscopic particles.

(56) The hybrid power electronic substrate **58** shown in FIGS. **27-30**, due to the lack of a direct copper-to-ceramic bond, eliminates the need for the high temperature bonding processes described above. In addition, because there is no need for a high temperature bonding or other sintering process, the substrate **58** including a ceramic layer can be formed using laminating or other pressure bonding processes.

(57) Implementations of IMS panels prior to singulation may have sizes of, or of about, 1 square meter, and may have the form of a square or of a rectangle. Implementations of DBC substrate panels prior to singulation may be wafer-shaped and may have sizes of, or of about, 5 inches by 7 inches.

(58) Implementations of power electronic substrates disclosed herein may be used, by non-limiting example, as substrates for insulated gate bipolar transistor (IGBT) power modules, intelligent power modules (IPMs), power integrated modules (PIMs), power metal-oxide-semiconductor field-effect-transistors (MOSFETs), and the like. In implementations terminals of a semiconductor package may be formed of the copper layers described herein. Packages formed using the power electronic substrates disclosed herein may include top leads, side leads, down leads, glass to metal seals, surface mounts, liquid cooling, and the like.

(59) PIM products may use DBC substrates with thicker copper trace thicknesses while IPM products may use IMS substrates with thinner copper trace thicknesses. Thinner copper traces are better for fine line space for routing while thicker copper traces are better for thermal and electrical performance for power electronic devices. In implementations the power electronic substrates disclosed herein may allow both of these advantages to be realized on a single substrate. In such implementations the thicker copper traces are used for power lines for power electronics while the thinner copper traces may be used for the rest of the circuitry with fine line spacing, and/or for fine pitch circuitry, such as for one or more drivers. The use of some thinner copper traces may reduce overall substrate stress.

(60) In particular implementations a leadframe of a power electronic device may be bonded to the top layer (copper or nickel) of a power electronic substrate described herein. This may be done, in implementations, using a solder, such as by non-limiting example an Sn/Ag/Cu solder.

(61) As may be envisioned, the process of forming an IMS shown in FIG. **17** may be followed up by additional steps to form a stacked IMS. By non-limiting example, a second dielectric layer may be laminated over the traces (and nickel plating, if present) and a second copper layer (having a pattern therein, or not) may then be coupled to the second dielectric layer, with traces later formed in the second copper layer to form the stacked IMS for a power electronic, these later traces having, if desired, multiple thicknesses as previously described with respect to other traces.

(62) Implementations of substrates disclosed herein may utilize principles disclosed in U.S. Pat. No. 7,078,797 listing as inventors Suzuki et al., issued Jul. 18, 2006, titled "Hybrid Integrated Circuit Device," the disclosure of which is hereby entirely incorporated herein by reference. Furthermore, forming ground connections to substrates as illustrated in that reference, such as, by non-limited example shown in FIG. 1B of that reference, may be incorporated into power electronic substrate designs disclosed herein. Forming such connections may be accomplished, by

non-limited example, by etching or otherwise forming a through-hole through the dielectric material, ceramic layer, or other insulative layer during processing, using methods disclosed herein, and then coupling an electrical contact on a surface of a die to a grounded metallic baseplate using a wirebond or the like.

(63) Furthermore, substrate implementations like those disclosed herein by use the principles disclosed in U.S. Pat. No. 7,102,211, listing as inventors Ochiai et al., issued Sep. 5, 2006, titled "Semiconductor Device and Hybrid Integrated Circuit Device," the disclosure of which is hereby entirely incorporated herein by reference. Implementations of power electronic substrates disclosed herein may be used to form hybrid integrated circuit (HIC) devices such as those disclosed in that reference. The "fused leads" of an HIC package as shown in that reference, such as by non-limiting example those shown in FIG. 6B (elements 54, 55) of that reference, may be formed of the same copper layer that is used to make the traces **20** described herein.

(64) Substrate implementations like those may be formed employing the principles disclosed in U.S. Pat. No. 7,521,290, listing as inventors Takakusaki et al., issued Apr. 21, 2009, titled "Method of Manufacturing Circuit Device," the disclosure of which is hereby entirely incorporated herein by reference. The methods disclosed therein of attaching a leadframe to multiple substrates (or in other words to a single panel containing multiple non-singulated substrates prior to singulation), to then be singulated, such as by non-limiting example the elements shown in FIG. 3A of that reference, may be incorporated in and/or used together with power electronic devices disclosed herein.

(65) Implementations of substrates like those disclosed herein may be formed using the principles disclosed in U.S. Pat. No. 7,935,899, listing as inventors Takakusaki et al., issued May 3, 2011, titled "Circuit Device and Method of Manufacturing the same," the disclosure of which is hereby entirely incorporated herein by reference. Furthermore, packaging multiple HIC substrates within a single package as disclosed in that reference, such as that shown by non-limiting example in FIG. 1B and described in the specification of that reference, may be accomplished in part by forming several power electronic substrates according to methods disclosed herein in a single panel and then singulating each individual power electronic substrate, such as through punch or saw singulation, and interconnecting die and other components between HIC modules as shown in FIG. 1B of that reference.

(66) In various implementations of substrates disclosed herein, the principles disclosed in U.S. Pat. No. 8,448,842, listing as inventor Wu, issued May 28, 2013, titled "Advanced copper bonding (ACB) with ceramic substrate technology," may be employed, the disclosure of which is hereby entirely incorporated herein by reference. Any of the bonding techniques disclosed therein with respect to bonding copper layers to ceramic layers may be utilized in forming power electronic substrates disclosed herein including, by non-limiting example: forming a copper film having a thickness of less than 1 micron on a ceramic substrate by sputtering deposition under 0.00133 torr and 150 degrees Celsius; plating a copper layer of 10-50 microns at room temperature, and; bonding a copper foil to the ceramic substrate by diffusion bonding under environments of high temperature, vacuum, and negative pressure inertia gas or H.sub.2 partial pressure. In implementations a copper layer may be bonded to an aluminum oxide ceramic layer using methods described herein by heating in a sintering furnace up to 1000 degrees Celsius (or higher, such as about 1060 to about 1080 degrees Celsius) to form the eutectic layer described previously. In implementations no sputtering of copper onto a ceramic layer is needed to form the copper layer.

(67) Implementations of substrates disclosed herein that include a nickel layer may employ the methods and principles disclosed in U.S. Pat. No. 7,936,569, listing as inventors Takakusaki et al., issued May 3, 2011, titled "Circuit Device and Method of Manufacturing the same," the disclosure of which is hereby entirely incorporated herein by reference. Furthermore, any of the elements therein describing nickel plating over copper traces, heat sink elements, and other elements used when attaching a die to a copper trace and/or electrically coupling an electrical contact on the die with one or more traces, such as by non-limiting example the elements shown in FIG. 1C of that

reference and related description in the specification thereof, may be incorporated and/or used together with power electronic substrates disclosed herein. Additionally, insulating layers and/or dielectric layers described herein may include any of the elements, characteristics, features and the like of resins and/or insulating layers described in U.S. Pat. No. 7,936,569.

(68) Implementations of substrates like those disclosed herein may employ the principles disclosed in Japan Patent Application Publication No. JP-2006-237561, listing as inventors Takakusaki et al., published Sep. 7, 2006, titled "Circuit Device and its Manufacturing Process," the disclosure of which is hereby entirely incorporated herein by reference. Furthermore, any of the elements therein that disclose nickel plating over copper traces, heat sink elements, and other elements used when attaching a die to a copper trace and/or electrically coupling an electrical contact on the die with one or more traces, such as by non-limiting example the elements shown in FIG. 1C of that reference and related description in the specification thereof, may be incorporated and/or used together with power electronic substrates disclosed herein. Additionally, insulating layers and/or dielectric layers described herein may include any of the elements, characteristics, features and the like of resins and/or insulating layers described in U.S. Pat. No. 7,936,569 previously incorporated by reference.

(69) Implementations of substrates like those disclosed herein may be manufactured using the principles disclosed in Japan Patent Application Publication No. JP-2008-022033, listing as inventors Mizutani et al., published Jan. 31, 2008, titled "Hybrid Integrated Circuit Device," the disclosure of which is hereby entirely incorporated herein by reference. Furthermore, any of the v-score techniques applied to the substrates as disclosed therein in at least FIGS. 6-8 and 10, and related disclosure in the specification thereof, may be applied to and/or used with power electronic substrates disclosed herein to aid with singulation. In implementations such v-scores may be applied to the metallic baseplates described herein. In implementations double v-scores may be utilized wherein a plurality of v-scores are on an underside of the metallic baseplate and a corresponding plurality of v-scores are on the upper side of the metallic baseplate and aligned with the v-scores on the underside of the metallic baseplate to aid with singulation.

(70) In places where the description above refers to particular implementations of substrate structures and methods of manufacture and implementing components, sub-components, methods and sub-methods, it should be readily apparent that a number of modifications may be made without departing from the spirit thereof and that these implementations, implementing components, sub-components, methods and sub-methods may be applied to other substrate structures and methods of manufacture.

Claims

1. A power electronic substrate, comprising: a plurality of copper traces coupled to a first dielectric layer, wherein the plurality of copper traces comprise a first thickness and a second thickness measured perpendicularly to the first dielectric layer; a ceramic layer comprising a first side and a second side, wherein the first side of the ceramic layer is coupled to the first dielectric layer; a second dielectric layer coupled to the second side of the ceramic layer; and a metallic baseplate coupled to the second dielectric layer; wherein the ceramic layer comprises two different thicknesses, a first surface of the ceramic layer comprising a step therein; wherein the first dielectric layer covers a first portion of the first surface of the ceramic layer and a second portion of the first surface of the ceramic layer; and wherein the step in the first surface separates the first portion from the second portion.
2. The substrate of claim 1, wherein the first dielectric layer and the second dielectric layer comprise one of an epoxy or a resin.
3. The substrate of claim 1, wherein the first dielectric layer and the second dielectric layer are between 25-300 micrometers thick.

4. The substrate of claim 1, wherein the plurality of copper traces further comprises a third thickness different in magnitude from the first thickness and the second thickness, the third thickness measured perpendicularly to the first dielectric layer.
5. The substrate of claim 1, further comprising a plurality of nickel traces plated onto the plurality of copper traces.
6. The substrate of claim 1, further comprising a third dielectric layer coupled to the plurality of copper traces and a second plurality of copper traces coupled to the third dielectric layer.
7. The substrate of claim 1, further comprising bonding patterns in one of the first dielectric layer, the second dielectric layer, or both the first dielectric layer and the second dielectric layer.
8. The substrate of claim 7, wherein the bonding patterns comprise one of bonding ridges, conical projections, or pyramidal projections.
9. A power electronic substrate, comprising: a plurality of metallic traces coupled to a first dielectric layer, wherein the plurality of metallic traces comprise a first thickness and a second thickness measured perpendicularly to the first dielectric layer; a ceramic layer comprising a first side and a second side, wherein the first side of the ceramic layer is coupled to the first dielectric layer; a second dielectric layer coupled to the second side of the ceramic layer; and a metallic baseplate coupled to the second dielectric layer; wherein the ceramic layer comprises a recess formed in the first side; and wherein a portion of the plurality of metallic traces are coupled within the recess.
10. The substrate of claim 9, further comprising bonding patterns in one of the first dielectric layer, the second dielectric layer, and both the first dielectric layer and the second dielectric layer.
11. The substrate of claim 10, wherein the bonding patterns comprise one of bonding ridges, conical projections, and pyramidal projections.
12. The substrate of claim 9, wherein the first dielectric layer and the second dielectric layer are between 25-300 micrometers thick.
13. The substrate of claim 9, wherein the plurality of metallic traces further comprises a third thickness different in magnitude from the first thickness and the second thickness, the third thickness measured perpendicular to the first dielectric layer.
14. The substrate of claim 9, further comprising a plurality of nickel traces plated onto the plurality of metallic traces.
15. The substrate of claim 9, further comprising a third dielectric layer coupled to the plurality of metallic traces and a second plurality of metallic traces coupled to the third dielectric layer.
16. A power electronic substrate, comprising: a plurality of metallic traces coupled to a first dielectric layer, wherein the plurality of metallic traces comprise a first thickness and a second thickness measured perpendicularly to the first dielectric layer; a ceramic layer comprising a first surface and a second surface, wherein the first surface of the ceramic layer is coupled to the first dielectric layer; a second dielectric layer coupled to the second surface of the ceramic layer; and a metallic baseplate coupled to the second dielectric layer; wherein the ceramic layer comprises two different thicknesses, the first surface of the ceramic layer comprising a step therein; wherein the first dielectric layer covers a first portion of the first surface of the ceramic layer and a second portion of the first surface of the ceramic layer; wherein the step in the first surface separates the first portion from the second portion; and wherein a metallic trace of the plurality of metallic traces overlies the step.
17. The substrate of claim 16, wherein the plurality of metallic traces further comprises a third thickness different in magnitude from the first thickness and the second thickness, the third thickness measured perpendicularly to the first dielectric layer.
18. The substrate of claim 16, further comprising a plurality of nickel traces plated onto the plurality of metallic traces.
19. The substrate of claim 16, further comprising bonding patterns in one of the first dielectric layer, the second dielectric layer, or both the first dielectric layer and the second dielectric layer.

20. The substrate of claim 19, wherein the bonding patterns comprise one of bonding ridges, conical projections, or pyramidal projections.
